

ML Scan Parameters

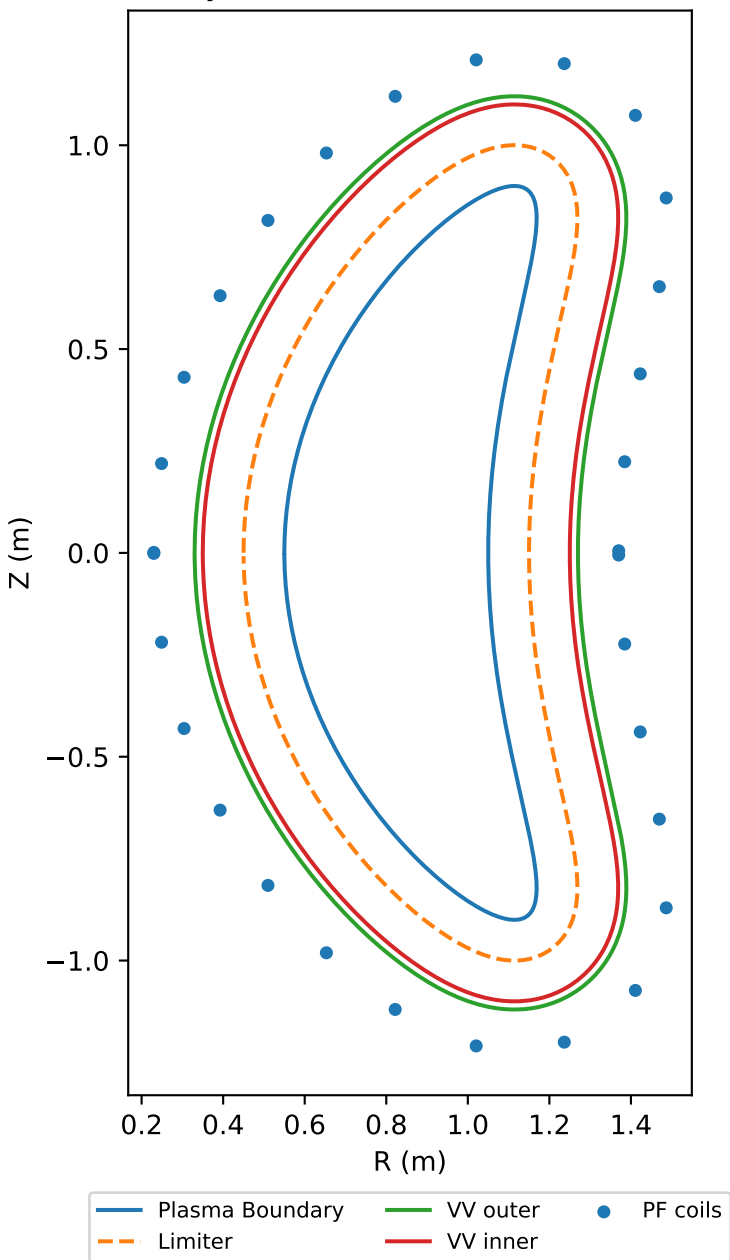
C-shape plasma $(r,z) = (A + B*\sin t + C*\cos 2t, F + G*\cos t - H*\sin 2t)$

A=0.9600 B=0.2500 C=0.1600
G=0.9000 H=0.0100 F=0.0
R0=0.8000 m kappa_target=3.600 delta=-0.010
B0=1.8 T Ip=1.00 MA

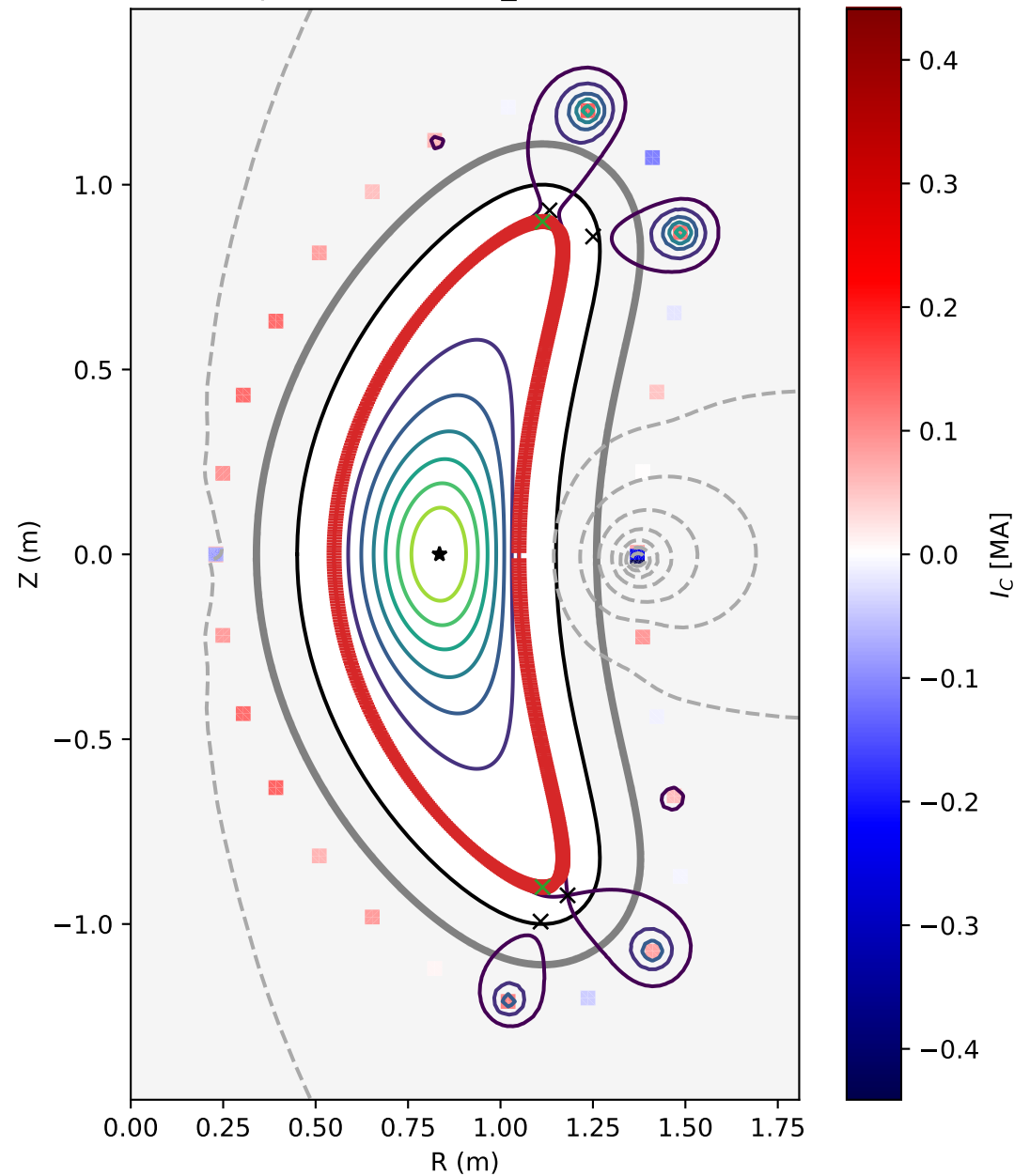
Plasma profiles:
pp_prof: $(1 - \psi^{2.120})^{0.530}$
ffp_prof: $(1 - \psi^{1.024})^{2.882}$

Scan params: B=2.5 C=16 G=9 H=1
alpha_pp=2.120 gamma_pp=0.530
alpha_ff=1.024 gamma_ff=2.882

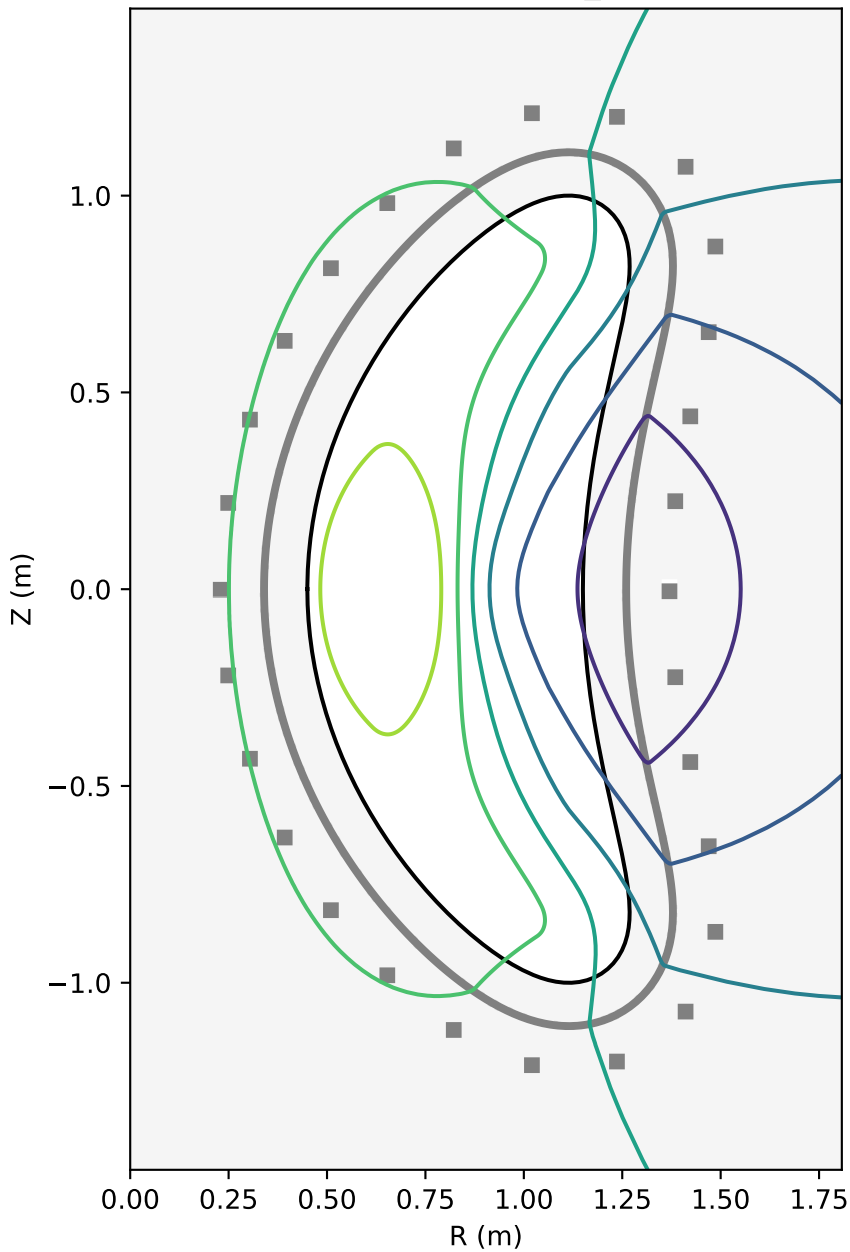
Geometry B=2.5, C=16, G=9, H=1, $\kappa=3.6$



Equilibrium flux $\kappa_{\text{actual}}=3.00$



Unstable eigenmode $\gamma\tau_w = -0.353$



Vertical Stability Analysis

Vertical Stability Results

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B=2.5 C=16 G=9 H=1
alpha_pp=2.120 gamma_pp=0.530
alpha_ff=1.024 gamma_ff=2.882

kappa_target = 3.600
kappa_actual = 3.000
beta_N = 1.040 %

growth rate (gamma) = -9.7934E+01 s-1
wall time (tau_w) = 3.6076E-03 s
gamma * tau_w = -0.3533
max coil current = 0.441 MA
Category: passive stable

axis_offset_pct = 4.38%

Equilibrium Statistics

Equilibrium Statistics:	
Topology	= Diverted
Toroidal Current [A]	= 1.0000E+06
Current Centroid [m]	= 0.850 0.000
Magnetic Axis [m]	= 0.835 0.000
Elongation	= 3.000 (U: 2.886, L: 3.114)
Triangularity	= -0.951 (U: -0.778, L: -1.123)
Plasma Volume [m^3]	= 4.043
q_0, q_95	= 0.512 5.340
Peak Pressure [Pa]	= 1.2624E+05
Stored Energy [J]	= 1.4365E+05
<Beta_pol> [%]	= 58.2162
<Beta_tor> [%]	= 2.0677
<Beta_n> [%]	= 1.0396
Diamagnetic flux [Wb]	= 9.0620E-03
Toroidal flux [Wb]	= 1.2368E+00
l_i	= 1.1864